

Astra Campus OS - Project Synopsis

1. Project Title

Astra Campus OS – AI-Powered Smart Campus Infrastructure Management System

2. Introduction

Astra Campus OS is an AI-powered platform that digitizes campus maintenance, complaint management, and infrastructure monitoring using automation and intelligent workflows.

3. Problem Statement

Traditional campus maintenance relies on manual complaint handling, delayed communication, and inefficient tracking, leading to slower resolution and poor resource utilization.

4. Proposed Solution

A centralized AI-driven platform enabling complaint registration via text, voice, image, or QR code, with automatic categorization, prioritization, and technician assignment.

5. Objectives

- Digitize maintenance workflows.
- Reduce complaint resolution time.
- Automate complaint classification using AI.
- Enable QR-based asset tracking.
- Provide analytics and dashboards.

6. Key Features

AI Complaint Classification, Voice Complaint Registration, QR Asset Tracking, Predictive Maintenance, Role-Based Dashboards, Real-Time Analytics, Notifications, Ticket Lifecycle Management.

7. Technology Stack

Frontend: React, TypeScript, Tailwind CSS, Framer Motion

Backend: Node.js, Express.js

Database: MongoDB

AI: Gemini/OpenAI APIs

Authentication: Firebase/Clerk

8. Working

Student submits complaint → AI analyzes severity → Ticket assigned automatically → Warden reviews → Technician resolves → Dashboard updates in real time.

9. Expected Outcomes

Faster maintenance, improved transparency, predictive maintenance, better resource allocation, and enhanced campus operations.

10. Future Scope

IoT integration, mobile application, predictive analytics, digital twin visualization, smart energy management.

11. Conclusion

Astra Campus OS is a scalable AI-powered platform that modernizes campus infrastructure management through automation, analytics, and intelligent decision-making.